

$$\begin{aligned}
s_{t+i} &\sim P\left(\cdot \mid s_{t+i-1}, a_{t+i-1}^{\text{prev}}\right), \\
z_{t+i} &= E(\mathcal{O}(s_{t+i})) \quad , \quad (i, z_{t+i}) \in \mathcal{F}_{t,k}
\end{aligned}$$